At equilibrium, $\begin{pmatrix} S & H \end{pmatrix}\begin{pmatrix} \frac{1}{3} & \frac{2}{3} \\ \frac{4}{5} & \frac{1}{5} \end{pmatrix} = \begin{pmatrix} S & H \end{pmatrix}$

where $S + H = 1$

$$\frac{1}{3}S + \frac{4}{5}H = S$$
$$\frac{1}{3}(1-H) + \frac{4}{5}H = 1\text{-}H$$

On solving this, we get

$$S = \frac{6}{11} \text{ and } H = \frac{5}{11}$$

In the long run, on a randomly selected day, his chances of being happy is $\frac{5}{11}$.

Example 1.27

Akash bats according to the following traits. If he makes a hit (S), there is a 25% chance that he will make a hit his next time at bat. If he fails to hit (F), there is a 35% chance that he will make a hit his next time at bat. Find the transition probability matrix for the data and determine Akash's long- range batting average.

Solution:

The Transition probability matrix is

$$T = \begin{pmatrix} 0.25 & 0.75 \\ 0.35 & 0.65 \end{pmatrix}$$

At equilibrium, $\begin{pmatrix} S & F \end{pmatrix}\begin{pmatrix} 0.25 & 0.75 \\ 0.35 & 0.65 \end{pmatrix} = \begin{pmatrix} S & F \end{pmatrix}$

where $S + F = 1$

$0.25\,S + 0.35\,F = S$

$0.25\,S + 0.35\,(1 - S) = S$

On solving this, we get, $S = \frac{0.35}{1.10}$

$$\Rightarrow S = 0.318 \text{ and } F = 0.682$$

$\therefore$ Akash's batting average is 31.8%

Example 1.28

80% of students who do maths work during one study period, will do the maths work at the next study period. 30% of students who do english work during one study period, will do the english work at the next study period.

Initially there were 60 students do maths work and 40 students do english work. Calculate,

(i) The transition probability matrix

(ii) The number of students who do maths work, english work for the next subsequent 2 study periods.

Solution

(i) Transition probability matrix

$$T = \begin{matrix} \\ M \\ E \end{matrix}\begin{matrix} \begin{matrix} M & E \end{matrix} \\ \begin{pmatrix} 0.8 & 0.2 \\ 0.7 & 0.3 \end{pmatrix} \end{matrix}$$

After one study period,

$$\begin{matrix} M & E \end{matrix} \quad \begin{matrix} M & E \end{matrix} \quad \begin{matrix} M & E \end{matrix}$$
$$\begin{pmatrix} 60 & 40 \end{pmatrix} \quad \begin{matrix} \\ M \\ E \end{matrix}\begin{matrix} \begin{matrix} M & E \end{matrix} \\ \begin{pmatrix} 0.8 & 0.2 \\ 0.7 & 0.3 \end{pmatrix} \end{matrix} = \begin{pmatrix} 76 & 24 \end{pmatrix}$$

So in the very next study period, there will be 76 students do maths work and 24 students do the English work.

After two study periods,

$$\begin{matrix} M & E \end{matrix} \quad \begin{matrix} M & E \end{matrix}$$
$$\begin{pmatrix} 76 & 24 \end{pmatrix} \quad \begin{matrix} \\ M \\ E \end{matrix}\begin{matrix} \begin{matrix} M & E \end{matrix} \\ \begin{pmatrix} 0.8 & 0.2 \\ 0.7 & 0.3 \end{pmatrix} \end{matrix}$$
$$= \begin{pmatrix} 60.8 + 16.8 & 15.2 + 7.2 \end{pmatrix}$$
$$= \begin{pmatrix} 77.6 & 22.4 \end{pmatrix}$$

After two study periods there will be 78 (approx) students do maths work and 22 (approx) students do English work.

Aliter

$$\begin{matrix} M & E \end{matrix} \quad \begin{matrix} M & E \end{matrix}$$
$$\begin{pmatrix} 60 & 40 \end{pmatrix} \quad \begin{matrix} \\ M \\ E \end{matrix}\begin{matrix} \begin{matrix} M & E \end{matrix} \\ \begin{pmatrix} 0.8 & 0.2 \\ 0.7 & 0.3 \end{pmatrix}^2 \end{matrix}$$

$$\begin{matrix} M & E \end{matrix} \quad \begin{matrix} M & E \end{matrix}$$
$$= \begin{pmatrix} 60 & 40 \end{pmatrix} \quad \begin{matrix} \\ M \\ E \end{matrix}\begin{matrix} \begin{matrix} M & E \end{matrix} \\ \begin{pmatrix} 0.78 & 0.22 \\ 0.77 & 0.23 \end{pmatrix} \end{matrix}$$

$= (46.8 + 30.8 \quad 13.2 + 9.2)$

$= (77.6 \quad 22.4)$

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